## MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, TRAVELING WAVE, CW AMPLIFIER TYPE 8551[1]

*The complete requirements for procuring the electron tube described herein shall consist of this document and the issue in effect Specification MIL-E-1.*

**DESCRIPTION:** 1-Watt Input, Periodic Permanent-Magnet Focused, Coaxial Coupling, Operation Range 4400 Mc to 5000 Mc.

**RATINGS:** (Notes 1 and 4).

| Parameter: | $E_f$ | $E_c$ | $I_c$ | $I_k$ | $E_b$ | $I_b$ | $E_w$ | $I_w$ | $P_{i(r-f)}$ | $P_p$ | $t_k$ | $i_{f(surge)}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | mAdc | mAdc | Vdc | mAdc | Vdc | mAdc | W | W | sec | A |
| Maximum: | 7.35 | 900 | 1.0 | 25 | 1800 | 25 | 1800 | 2.5 | 1.0 | 40 | ..... | 6.0 |
| Minimum: | 6.65 | ..... | ..... | ..... | ..... | ..... | ..... | ..... | ..... | ..... | 180 | ..... |

**TEST CONDITIONS:** (Note 5).

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| (1) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| (2) | 7.0 | (Note 6) | (Note 6) | (Note 6) | (Note 6) | (Note 6) | (Note 6) | (Note 6) |
| | as req'd | 0 | ..... | ..... | | | | |
| | as req'd | ..... | 180 | ..... | | | | |

**OUTLINE AND DIMENSIONS:** (See fig. 1.)

**RF INPUT CONNECTOR:** TNC Female.

**RF OUTPUT CONNECTOR:** TNC Female.

**NOMINAL RF INPUT AND OUTPUT IMPEDANCE:** 50 ohms.

**VOLTAGE REFERENCE:** Cathode. (Collector connected internally to capsule; capsule to be grounded for operation).

**COOLING:** Forced air, at rate of 5 CFM minimum through capsule.

**BASE (DC) CONNECTOR:** Winchester Type PM6P LSH, or equal.

**WEIGHT:** 5.25 lbs., max.

[1] Previously known as TWT Type HA-85N.

FSC 5960

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**MOUNTING POSITION:** Any.

**OPERATING ALTITUDE:** 40,000 ft., max.

**DC POWER CONNECTOR TERMINALS DATA:** (See Fig. 1.)

| *Pin* | *Element* |
|---|---|
| A | Cathode. |
| B | Anode. |
| C | Helix. |
| D | Heater. |
| E | Heater. |
| F | Capsule (ground). |



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| MIL-E-1 Method (per app'x. E) or reference paragraph | Requirement or test | Conditions | AQL (percent defective) | Inspection level or code | Symbol | Limits | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Minimum | Maximum | Unit |
| | *General* | | | | | | | |
| 3.2 | Qualification | Required; (Note 7) | | | | | | |
| 3.1.1 | Performance | (Note 8) | | | | | | |
| 3.7 | Marking | (Notes 2, 9) | | | | | | |
| E 50.2 | Holding period | t = 24 hrs, min; (Note 6). | | | | | | |
| D 20 | Dimensions | (Note 10) | | | | | | |
| | Shelf life | (Note 11) | | | | | | |
| | **QUALITY CONFORMANCE INSPECTION, PART 1 — (Note 12)** | | | | | | | |
| 1261 | Heater voltage | $I_1$ = 1.1 max (Note 6) | 0.65 | II | $E_f$ | 6.65 | 7.35 | V |
| 1261 | Anode voltage | $I_C$ = 0.1 mAdc max (Note 6). | 0.65 | II | $E_a$ | 800 | 700 | Vdc |
| 1261 | Helix voltage | $I_W$ = 2.0 mAdc max (Note 6). | 0.65 | II | $E_W$ | 1400 | 1600 | Vdc |
| 1261 | Collector voltage | $I_b$ = 20.0 mAdc max (Note 6). | 0.65 | II | $E_b$ | 1400 | 1600 V | Vdc |
| 1256 | Cathode current | (Note 6a) | 0.65 | II | $I_k$ | | 20 | mAdc |
| | Small-signal gain | Test Cond. 2 (Notes 3, 13). | 0.65 | II | SSG | 8.5 | 4.5 | db |
| | Saturated power output | Test Cond. 2 (Notes 3, 14). | 0.65 | II | $P_o$ | 80 | | dbm |
| | Gain at 30 dbm $P_s$ | Test Cond. 2 (Notes 3, 15). | 0.65 | II | G | 80 | | db |
| | VSWR, input/output | Test Cond. 1 (Notes 3, 16). | 0.65 | II | ratio | | 2:1 | |
| | Stability | Test Cond. 2 (Notes 3, 17). | 0.65 | II | | | | |
| | **QUALITY CONFORMANCE INSPECTION, PART 2 — (Note 18)** | | | | | | | |
| | Vibration | No voltage applied (Note 19). | 6.5 | L6 | | | | |
| | Shock | No voltages (Note 20) | 6.5 | L6 | | | | |
| | Vibration and Shock test end-points. | (Note 21) | | | | | | |





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| MIL-E-1 Method (per app'x. E) or reference paragraph | Requirement or test | Conditions | AQL (percent defective) | Inspection level or code | Symbol | Limits | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Minimum | Maximum | Unit |
| | **QUALITY CONFORMANCE INSPECTION, PART 3, LIFE — (Note 22)** | | | | | | | |
| | Operation Life (1000 hours). | Test Cond. 2 (Note 23) .... | | | | | | |
| 4.7.3 | Operation Life test end points. | (Note 24) ..... | | | | | | |
| | **QUALITY CONFORMANCE INSPECTION, PART 3, PERIODIC — (Note 25)** | | | | | | | |
| | Barometric pressure, reduced (nonoperating). | Test Cond. 1; pressure — 140 mmHg, min; t — 70 minutes, min. | | | | | | |
| | Temperature cycling (nonoperating). | Test Cond. 1; T(low) — —55°C, max; T(high) — +72°C, min; (note 26). | | | | | | |
| | Temperature cycling (operating). | Test Cond. 2; T(low) — —40°C, max; T(high) — +55°C, min; (Note 27). | | | | | | |
| | Moisture resistance ..... | Test procedure per MIL-STD-202, Method 106, 14 cycles. | | | | | | |
| | *End-point tests for "Periodic" Group test samples.* | (Note 28) ..... | | | | | | |
| 1136 | Container drop ..... | Required; (Note 29) ..... | | | | | | |
| 5. | Preparation for delivery .... | (Note 30) ..... | | | | | | |

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**NOTES:**

1. These ratings are based upon the "absolute system" as discussed in Specification MIL-E-1, paragraph 6.5. Proper application and safeguard of the tube will best be served by strict compliance with the optimum electrical input conditions (see Note 2 below) and with the recorded performance data (see Note 3 below) and with the energization guidance (see Note 5 below) covered herein.

2. The following minimum amount of additional data: values for optimum service protection and operation of the tube, shall be marked on each tube using a suitable marking method conforming to requirements in Specification MIL-E-1, paragraph 3.7 (b) and 3.7 (c):
   (a) Heater voltage.
   (b) Helix voltage.
   (c) Collector voltage.
   (d) Cathode current.
   (e) Also, the RF "In" and "Out" connectors as shown in Fig. 1 herein.

3. At least 2 copies of a Tube Data Sheet shall be packaged with each tube shipped. At least one copy shall be a high-contrast, black-and-white copy, and at least one copy of the Tube Data Sheet shall be durably-coated but shall be flexible for convenient rolling into a small-diameter cylindrical form (to conserve packaging cubage), or the text on the required one, particular copy of the Tube Data Sheet shall be suitably imprinted (for durability) on a material that has the abovementioned flexibility properties. The Tube Data Sheet shall contain as a minimum, in addition to description information and connection-and-application guidances, a tabulation of the significant operational characteristics and limits (particularly, those characteristics referenced to this note) determined during actual testing of the individual tube. The tabulated data shall show values measured over the entire 4400 to 5000 Mc frequency band (see Note 5 herein) into a matched load, under indicated, applicable potentials. The Tube Data Sheet shall be of smallest dimensions practicable, except that all text matter contained thereon shall be imprinted in 8-print size type, minimum.

4. Symbols and abbreviations used herein are defined in Specification MIL-E-1 and as follows:

| | |
|---|---|
| Eb | collector voltage. |
| Ec | anode voltage. |
| Ew | helix voltage. |
| Ib | collector current. |
| Ic | anode current. |
| Iw | helix current. |
| $P_{i(r-f)}$ | input r-f power. |
| $P_c$ | collector (power) dissipation. |
| SSG | small-signal gain. |

5. Unless otherwise specified herein relative to particular tests all performance requirements shall be ascertained with the tube operating at each of the following frequency levels:

   F1: 4400 Mc
   F2: 4500 Mc
   F3: 4600 Mc
   F4: 4700 Mc
   F5: 4800 Mc
   F6: 4900 Mc
   F7: 5000 Mc

6. Energization of the tube for required performance tests, shall be in accordance with the following procedures and sequence:
   (a) Apply the rated or otherwise specified heater voltage for



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a minimum time period of 3 minutes. See (d) ,below.

(b) With the anode at rated or otherwise specified cathode potential, apply collector and helix voltages of equal value and within the range of 1500–1550 volts. See (d), below.

(c) Increase the anode voltage to produce the specified operating cathode current. If necessary, adjust helix and collector voltage (within their allowable voltage range) to establish and maintain helix current at less than 1.0 mA at the specified operation cathode current level.

(d) Electrical potentials, as present, may be applied simultaneously after the 3-minute heater warm-up period. **Caution**: Do not apply anode voltage without first or simultaneously applying collector and helix voltages. See (f), below.

(e) Optimize the helix and collector voltages. The values so determined shall, insofar as all specified performance requirements herein are met, be the values marked on the tube as per Note 2 above.

(f) For all tests, forced-air cooling at rate of 5 CFM, maximum through capsule may be applied.

7. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-1 whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Chief, Components and Materials Standardization Branch, Procurement and Production Directorate, ATTN: AMSEL-PP-E-CMD-2, Fort Monmouth, N. J. 07703.

8. All tests listed in Specification MIL-E-1, paragraph 3.1.1, are applicable except those per (therein-) referred paragraphs 3.6.2, App'x. E-40 and E-50.3.1 and E-1201, App'x. D-40 and D-20 (b) 2, App'x. E-1131. App'x. E-1216 and MIL-E-1 paragraph 3.5.2, App'x. E-1101 and E-1111 and E-1006 and E-40.1 (a) and (b), and App'x. D-70.2.

9. Type-designation marking of electron tubes procured on Department of Army contracts, and which have passed Government inspection and comply with all requirements of this specification, shall consist of: "USA-manufacturer's qualification code letters-tube designation." The letters "JAN" or any abbreviation thereof shall not be used. If any specification waiver has been granted, the combination "USA-manufacturer's qualification code letters" shall not be used to complete the type-designation marking.

10. The mounting-hole spacing dimensions shall be inspected as a Quality Conformance Inspection, Part I, test with an applicable AQL of 0.65%; all other dimensions shall be inspected under Quality Conformance Inspection, Part 2 (design) 6.5% AQL requirements.



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11. Tubes tested and procured under this specification sheet shall have a shelf life 12 months after date of shipment from manufacturer's plant. Tubes placed in operation during this period shall be capable of meeting all requirements herein. Tubes found to be defective within the shelf-life period shall be replaced by the manufacturer at no cost to the Government, or its contractors.

12. The AQL for the combined defects for attributes in Quality Conformance Inspection, Part 1, excluding mechanical, shall be 1.0%. A tube having one or more defects shall be counted as one defective.

13. Small-Signal Gain shall be measured in a test circuit conforming to figure 2 herein. Calibrate the signal-generator attenuator, and adjust the input signal for O dbm power output of the TWT. Small-Signal Gain can then be read directly from the signal-generator attenuator.

14. Saturated Power Output shall be measured in a test circuit conforming to figure 3 herein. Adjust the signal generator until maximum (saturated) power output is observed on the power meter. Adjust the variable attenuator for full-scale (O dbm) deflection on the power meter; then read power output ($P_o$) directly to the variable attenuator.

15. Gain at 30 dbm $P_o$ shall be measured in the test circuit conforming to Figure 4 herein. Calibrate the signal-generator attenuator, and adjust the input signal for 30 dbm power output of the TWT. Gain will equal the 30-dbm power output (value) minus the signal-generator attenuator reading.

16. The voltage standing wave ratio (VSWR) shall be measured using a swept-frequency technique, with a test circuit conforming to figure 4 herein. Adjust the sweep generator for a continuous swept output, over the frequency range 4400 to 5000 Mc. With a known load having a VSWR of 2:1 connected to the directional coupler, synchronize the oscilloscope with the sweep generator, and draw a line (temporary expedient mark) on the face of the oscilloscope indicating the detected broadband output level. Connect the TWT to the directional coupler, and observe the broadband output level as now indicated on the oscilloscope, this trace should be of equal or less amplitude than the originally marked output level, for the VSWR of 2:1 (or less).

17. With any load between open and short circuit on the input, and the output operating into a load having a VSWR of 3:1 or less, the tube shall be stable for all cathode currents within $\pm$ 1.0 mA of the specified and determined (see Notes 2 and 3 herein) cathode current value. As a minimum criteria for stablity, there shall be no evidence of discontinuity or "pip" on an oscilloscope trace.

18. Each tube in the respective sample subjected to the particular test in Quality Conformance Inspection Part 2 shall be individually identified, for test and evaluation purposes (see Note 21 herein) by a suitable marking or tagging method.

19. Test procedure in accordance with Method 204, Test Condition A, in Standard MIL-STD-202, except that acceleration = 8 G. minimum, shall be applicable during test.



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20. Shock test shall be performed using a Hyge "3-in. shock machine" as manufactured by Consolidated Engineering Corporation, or equivalent. Each tube under test shall be subjected to five blows in each of three mutually perpendicular directions provided that one direction shall be along the cylindrical axis of the tube; each blow shall be of 30 G acceleration and of 11 ± 1 millisecond duration; total-15 blows.

21. Following Vibration and Shock test, the respective tube samples shall meet all test requirements in Quality Conformance Inspection, Part 1 herein. The tubes shall then be further subjected to the Life Test specified in Quality Conformance Inspection Part 3, Life herein. Any failure (s) encountered for these particular (identified) tubes shall not be counted as a failure (s) against the separate sample accruing to the Life Test.

22. The following sampling plan and acceptance-ejection criteria shall apply to Life test:

    (a) Lot size: 50 tubes, maximum.
    Sample size: 5, selected at random.
    Acceptance number: 0 defectives.
    Rejection number: 1 defective.
    Where the first sample tested fails, the following shall apply for lot acceptance or rejection:
    Additional sample size: 10, selected at random.
    Acceptance number: 1 defectives, additional.
    Rejection number: 3 defectives (for both samples tested, combined).

    (b) Where a contract or order covers a total quantity of less than 50 tubes, either of the following sampling plans shall be selected and shall be effective without change of option for lot acceptance or rejection:
    Lot size: 20 (units), maximum.
    Sample size: 5, selected at random.
    Acceptance number: 0 defectives.
    Rejection number: 1 defective.
    Lot size: 20 (units), maximum.
    Sample size: 10, selected at random.
    Acceptance number: 1 defective.
    Rejection number: 2 defectives.

    (c) No "lot" of tubes shall be shipped until 1000-hour Life tests have been successfully completed for that lot. (See Note 23$_b$ herein.)

23. Life tests shall be performed at room-ambient temperature and shall consist of two test phases as follows:

    (a) With only heater voltage of 7.35 V applied to the tube, the heater shall be cycled on-and-off for a period of 100 hours. Each cycle shall be of 30 minutes duration, with a 20-minute "on" time and a 10-minute "off" time. At conclusion of the 100-hour cycling period, the heater voltage applied shall be



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reduced to normal: 7.0 V, and the following measurements shall be made:

Small-signal gain.
Saturated power output.
Anode voltage.
Cathode current.

(b) For the remaining Life Test hours (900), the tube shall be operated continuously with all applicable Test Condition 2 potentials applied. The measurements specified in (a) immediately above, shall be made at (approximately) 250, 500, 750, and 1000 hours.

24. The parametric values (test limits) specified in Quality Conformance Inspection Part 1 herein for the particular tests required in Note 23 (a) and (b) herein shall be met at conclusion of both Life Test "phases."

25. For this test group: "Part 3, Periodic," the requirements in Specification MIL–E–1, paragraph 4.1.1.2 shall apply except that report of any tube failures shall be directed to Procurement and Production Directorate, Components and Materials Division, ATTN: AMSEL–PP–E–CMD–2, Fort Monmouth, N.J. 07703.

26. Tubes shall be subjected for one cycle, as follows, to the temperature extremes specified. By means of a suitable thermocouple(s) attached to the capsule of the tube, stabilization of the tube at the temperature and for the period specified shall be ascertainable.

(a) 30 minutes at the low temperature specified.

(b) A transfer time, from one extreme temperature to the other extreme temperature, of ten minutes minimum.

(c) 30 minutes at the high temperature specified.

27. The following measurements shall be made immediately prior to the end of the 30-minute temperature-extreme subjections of the tube, and the test values found shall be within the limits specified for these tests in Quality Conformance Inspection Part 1 herein:

Small-signal gain.
Saturated power output.
Anode voltage.
Cathode current.

28. All sample tubes subjected to the tests in Quality Conformance Inspection, Part 3, Periodic herein shall be subjected, further, to the Life test specified in Quality Conformance Inspection, Part 3, Life herein and shall meet the pertinent Life-test end-points.

29. Not required for Qualification inspection.

30. Tubes shall be prepared for domestic or overseas shipment as specified in the contract or order.

31. Tubes subjected to the following destructive tests are not to be delivered on the contract or order:

Tests of Quality Conformance Inspection, Part 2 herein.

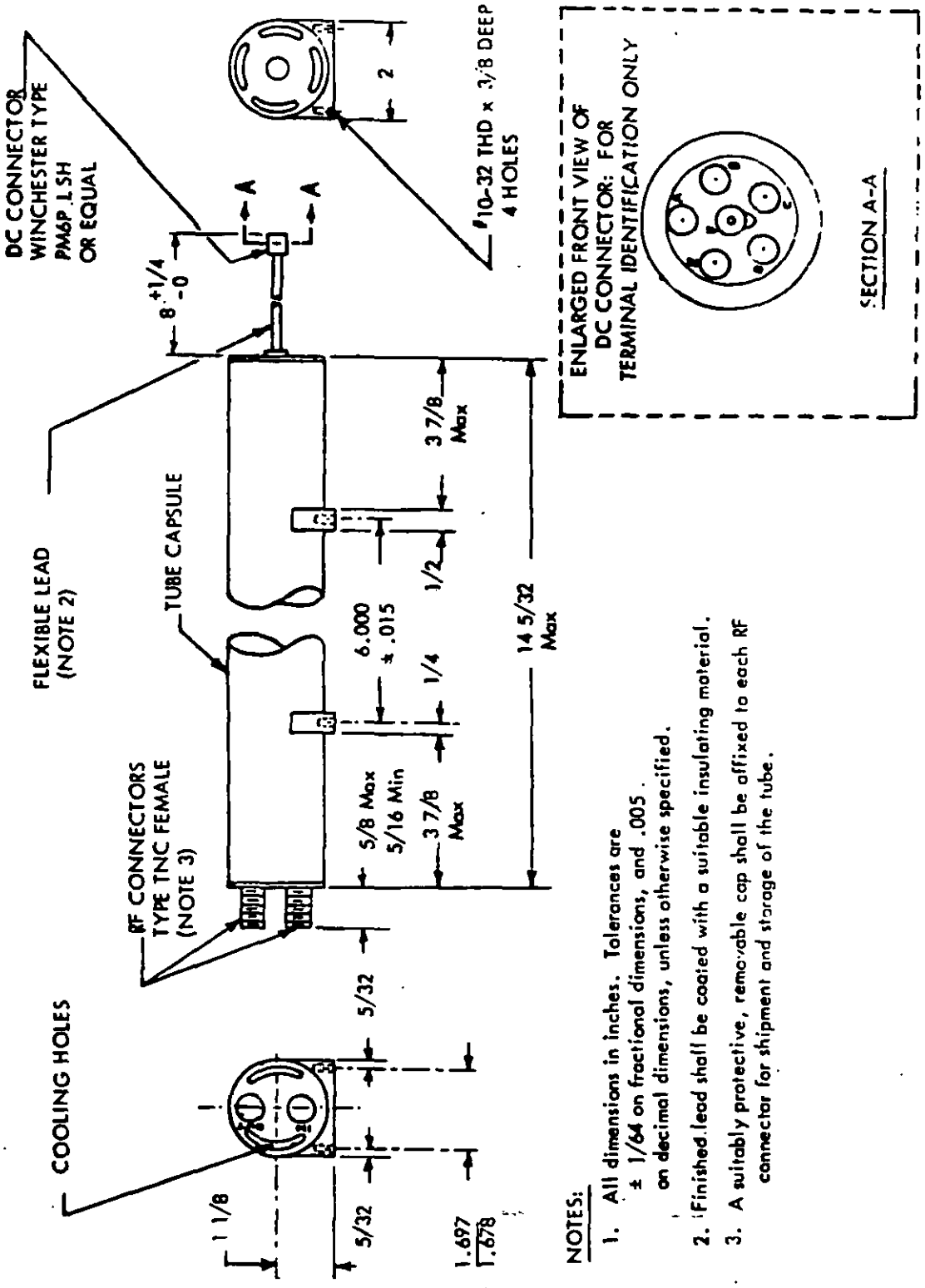
Tests of Quality Conformance Inspection, Part 3, Life and Part 3, Periodic herein.

Custodian:
Army—EL

Preparing activity:
Army—EL
Project No. 5960–A503



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NOTES:

1. All dimensions in inches. Tolerances are ± 1/64 on fractional dimensions, and .005 on decimal dimensions, unless otherwise specified.
2. Finished lead shall be coated with a suitable insulating material.
3. A suitably protective, removable cap shall be affixed to each RF connector for shipment and storage of the tube.

FIGURE 1. *Outline and dimensions.*



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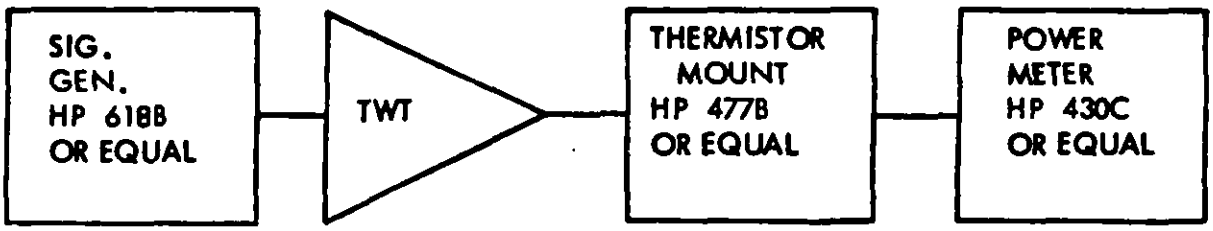



FIGURE 2. *Small-signal gain test circuit.*

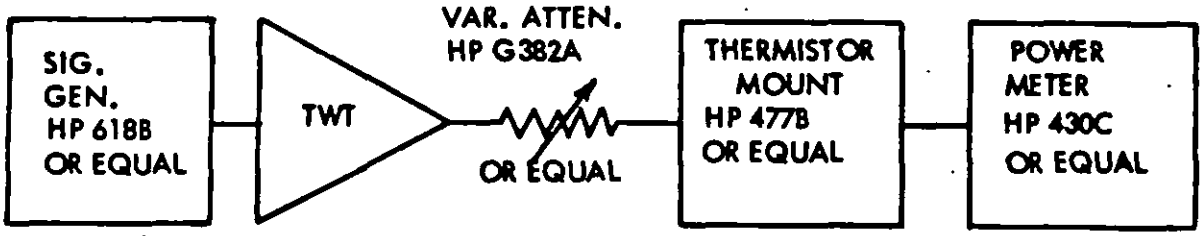



FIGURE 3. *Saturated power output test circuit.*

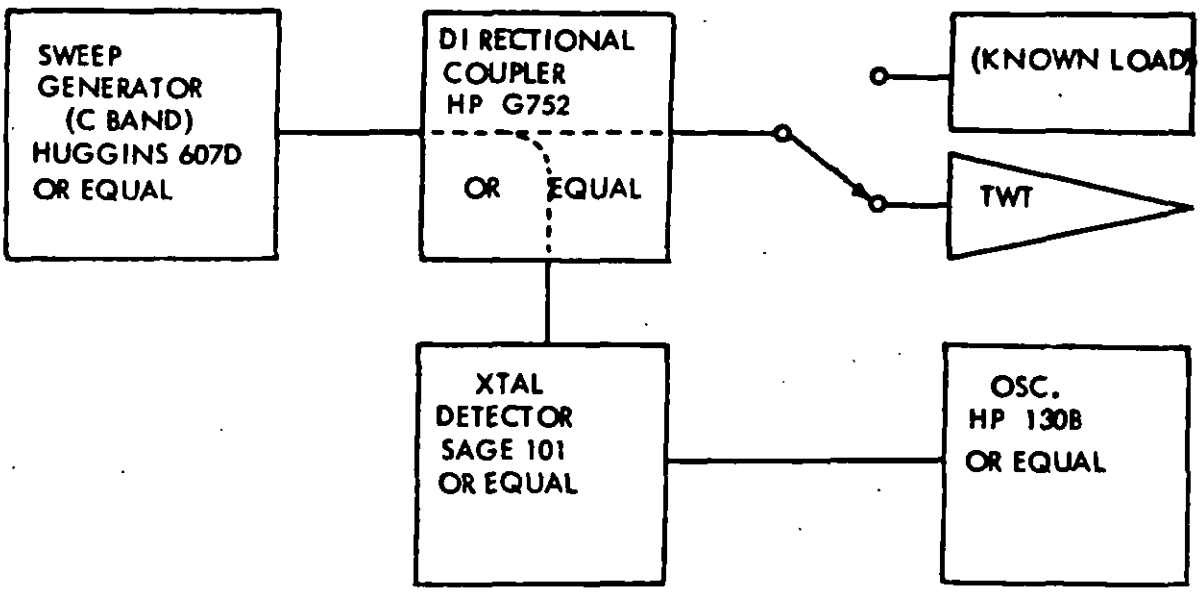



FIGURE 4. *VSWR, Input/output test circuit.*



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# SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

**INSTRUCTIONS**

*This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.*

| | | |
|---|---|---|
| SPECIFICATION | | |
| ORGANIZATION | | CITY AND STATE |
| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| | |
|---|---|
| SUBMITTED BY *(Printed or typed name and activity)* | DATE |

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